

Jargon buster

Before we delve into this FastTrack here is a lowdown on some of the more esoteric terms you are likely to encounter

CAGR: An abbreviation for compound annual growth rate, CAGR is an indication of the returns on an investment over a period of time. The CAGR is not a measurement of the actual returns, but a measure of the growth of an investment over a specified number of years, without taking into consideration the growth of the investment in each of those years. This is called a “smoothed annualised gain”. CAGR is calculated by using the formula $[(\text{Return}/\text{Investment})^{(1/\text{years})}] - 1$

IP: Intellectual Property are the assets created by the mind, and the laws in various nations that protect these assets. IP may or may not have a physical manifestation. Processes, which may be a trade practice or the designs of a machine, are also protected under IP laws. The copyright for works of art are also under the purview of IP.

SoC: System on a chip, a microchip design which fits in processor components, memory components of various types (RAM, ROM and Flash), various clocks and interface points for power supply and I/O devices, all on a single circuit board.

DFT design engineer: The Design for Test engineer has to devise test processes for prototype loadboards. These are circuits that resemble the chips, but are not manufactured in large volumes and are more costly. The Design for Test engineer, has to make sure that the chip is suitable for the particular application before it is fabricated.

EDA: Electronic Design Automation is the process of using software for laying out the design of a chip. The different components of a chip can be laid out by different EDA engineers separately, starting from a basic floor plan, then merged together in layers. Most modern EDA software allow for some kind of simulation based testing of the chip design.

VHDL: Very (High Speed Integrated Circuits) Hardware Description Language is a method initially used by the US defence services to abstract the lengthy and complex documentation of chips in simpler means. This allowed for useful and faster decisions based on chip design without detailed analysis of the implementation. VHDL is the industry's leading standard, although there are other hardware description languages.

MEEMS: Micro Electronics Electromechanical technologies use methods of chip fabrication for designing other kinds of electronic devices, which could be a radio or a light/solar panel. The advantage of MEEMS is that it is possible to make tiny, wireless devices that are low on power consumption.

Prior art: Prior art is information about a device, process or idea that has been made available to the public. A patent application is rejected, if the design of the patent was commonly available, or even known to members of the general public before the patent is applied for. The most popular example of this is Arthur C Clarke's thorough descriptions of satellites, due to which no one could patent them.